ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

Independent V3 design; 360–440 V DC input; hazardous non-isolated 200 V output.
Off-board choke, contactors and current sensor require the system assembly drawing.

SiC half bridge, DC link and output filter



文件: 01_stage.kicad_sch

High-side drive and short-circuit protection



文件: 02_hs.kicad_sch

Low-side drive and short-circuit protection



文件: 03_ls.kicad_sch

High-side isolated gate-bias supply



文件: 04_hs_bias.kicad_sch

Low-side isolated gate-bias supply



文件: 05_ls_bias.kicad_sch

Raw input, DC-link and output isolated voltage measurement



文件: 06_measure.kicad_sch

Independent isolated sensing supply



文件: 07_sense_bias.kicad_sch

Keyed control harness, fail-safe inputs and mounting



文件: 08_interface.kicad_sch

ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

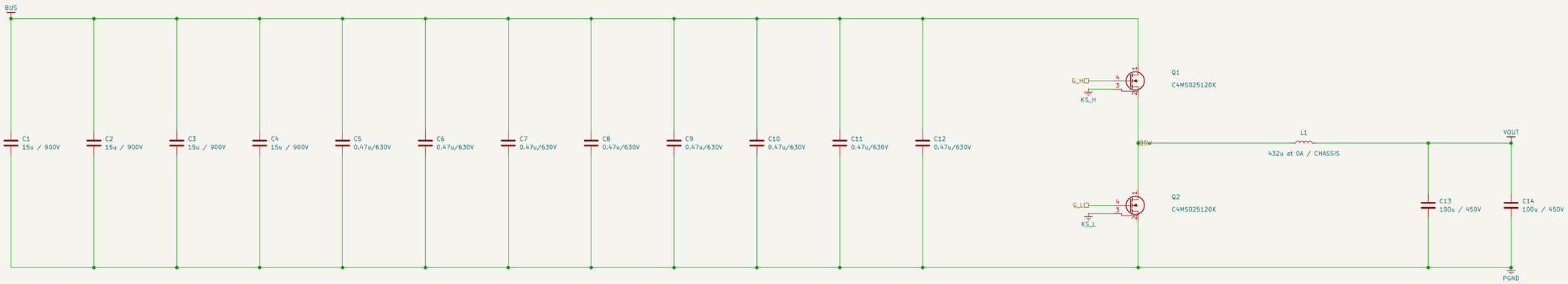
Sheet: /
File: sic-v3-power.kicad_sch

Title: V3 400V / 200V / 5kW — power

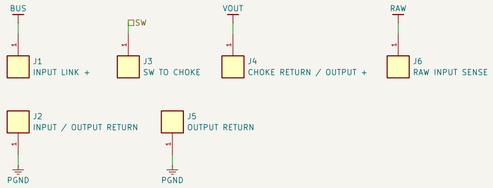
Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 1/9

SiC half bridge, DC link and output filter

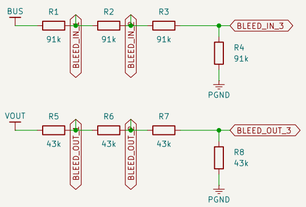
400 V DC link → vertical half bridge → external choke → 200 V output



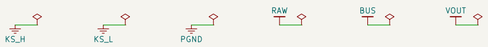
Chassis circuit connections — refer to system wiring project



Passive discharge independent of auxiliary power



Supply provenance (PWR_FLAG)



ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

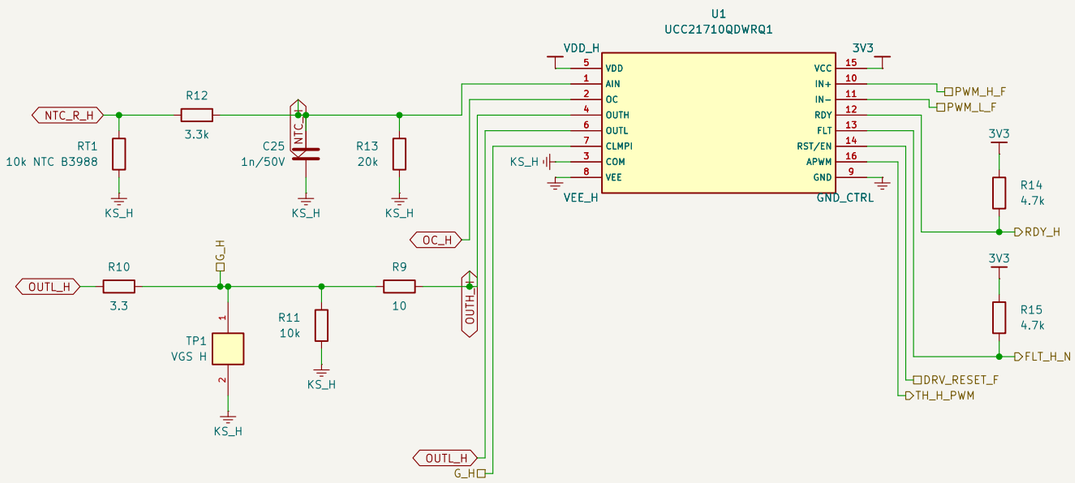
Sheet: /SiC half bridge, DC link and output filter/
File: 01_stage.kicad_sch

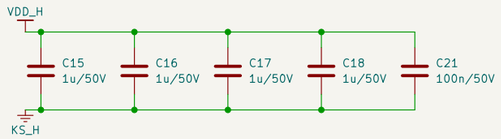
Title: SiC half bridge, DC link and output filter

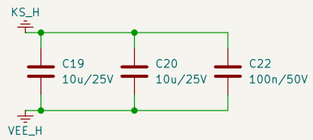
Size: A2	Date:	Rev: V3
KiCad E.D.A. 10.0.7		Id: 2/9

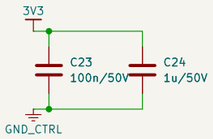
High-side drive and short-circuit protection

H: isolated gate driver, Kelvin return and clamp

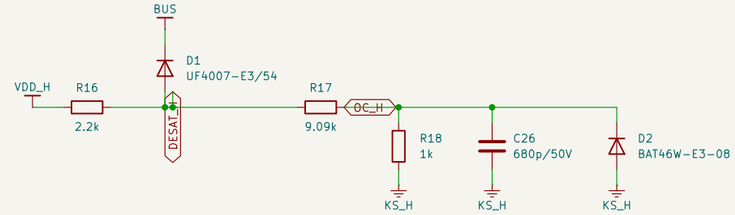








H: drain-voltage short-circuit detection



ENGINEERING PROTOTYPE REVIEW – RELEASE BLOCKED – NOT HARDWARE QUALIFIED

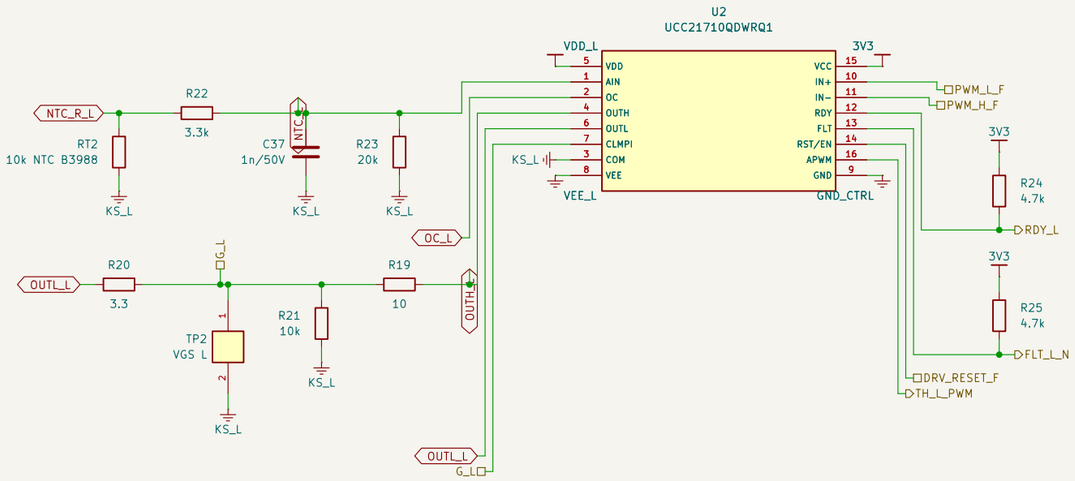
Sheet: /High-side drive and short-circuit protection/
File: 02_hs.kicad_sch

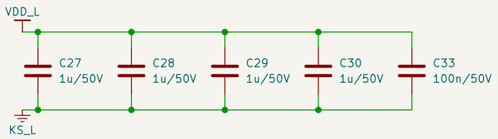
Title: High-side drive and short-circuit protection

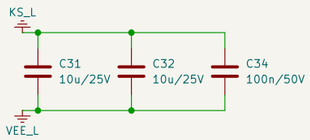
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KiCad E.D.A. 10.0.7		Id: 3/9

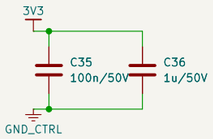
Low-side drive and short-circuit protection

L: isolated gate driver, Kelvin return and clamp

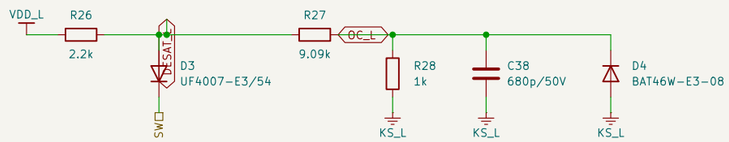








L: drain-voltage short-circuit detection



ENGINEERING PROTOTYPE REVIEW – RELEASE BLOCKED – NOT HARDWARE QUALIFIED

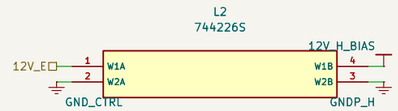
Sheet: /Low-side drive and short-circuit protection/
File: 03_ls.kicad_sch

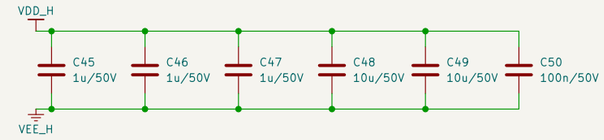
Title: Low-side drive and short-circuit protection

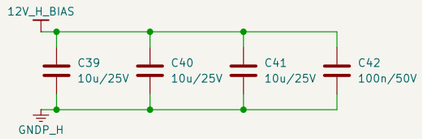
Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 4/9

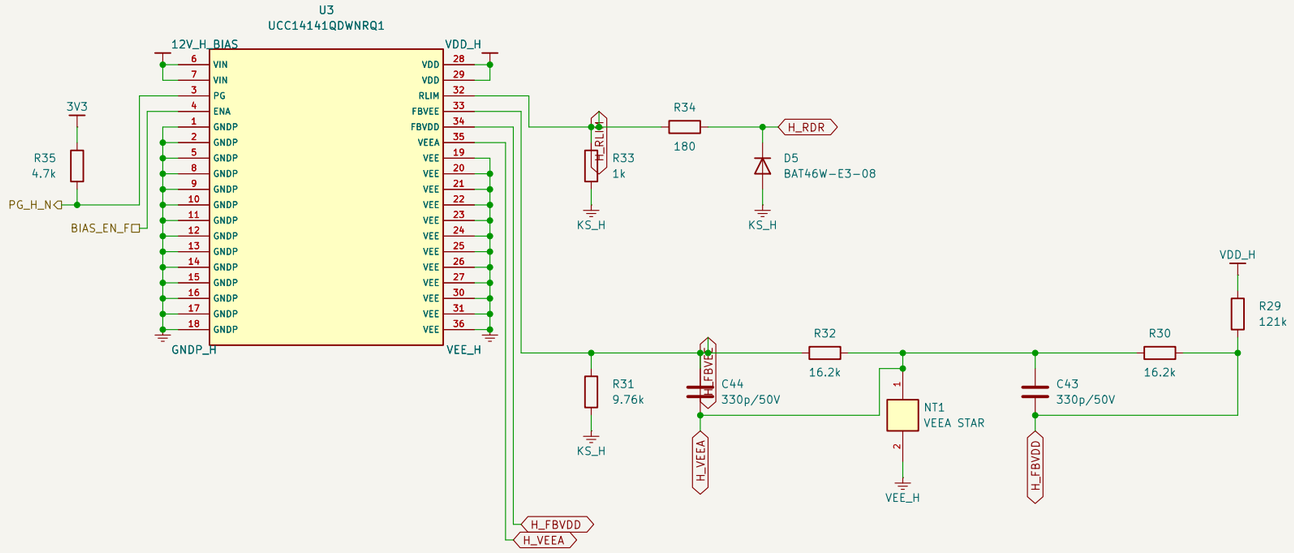
High-side isolated gate-bias supply

H: isolated DC/DC and physical feedback star

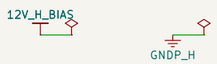








Supply provenance (PWR_FLAG)



ENGINEERING PROTOTYPE REVIEW – RELEASE BLOCKED – NOT HARDWARE QUALIFIED

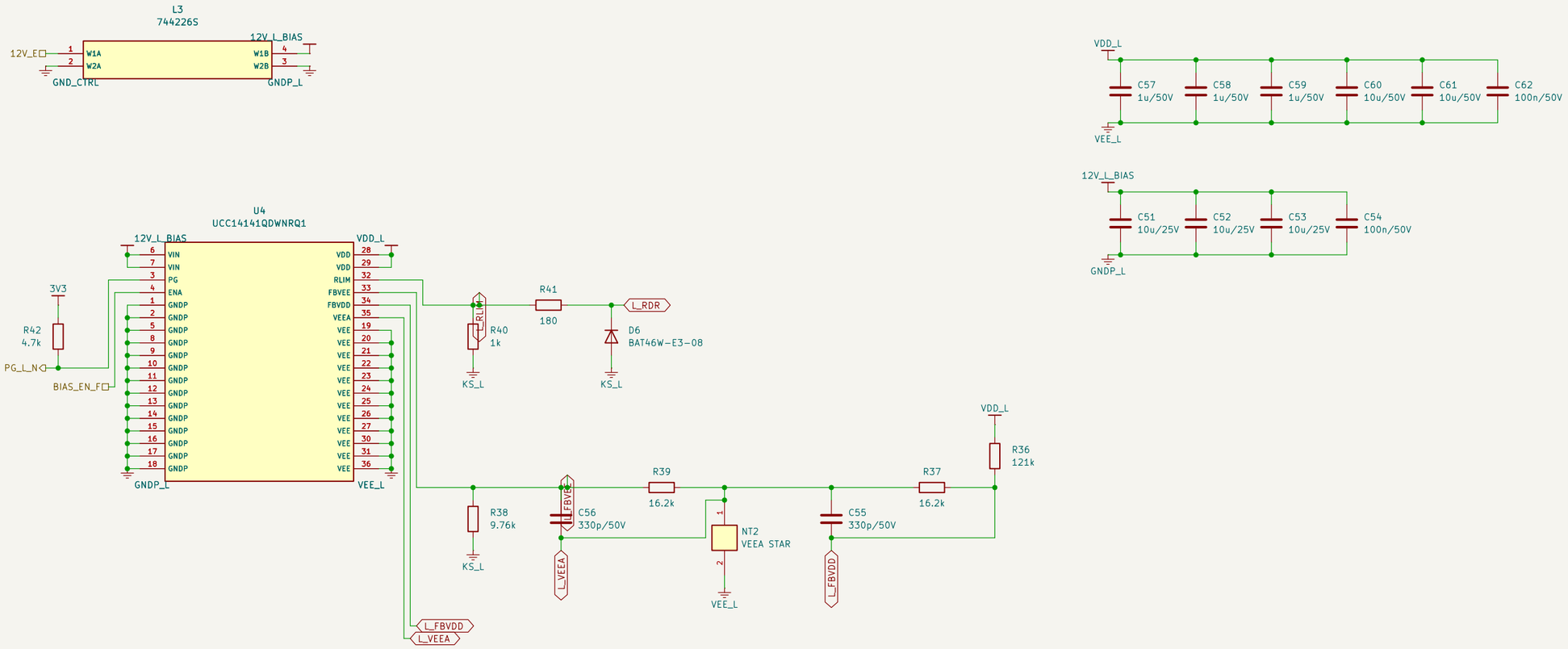
Sheet: /High-side isolated gate-bias supply/
File: 04_hs_bias.kicad_sch

Title: High-side isolated gate-bias supply

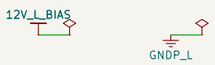
Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 5/9

Low-side isolated gate-bias supply

L: isolated DC/DC and physical feedback star



Supply provenance (PWR_FLAG)



ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

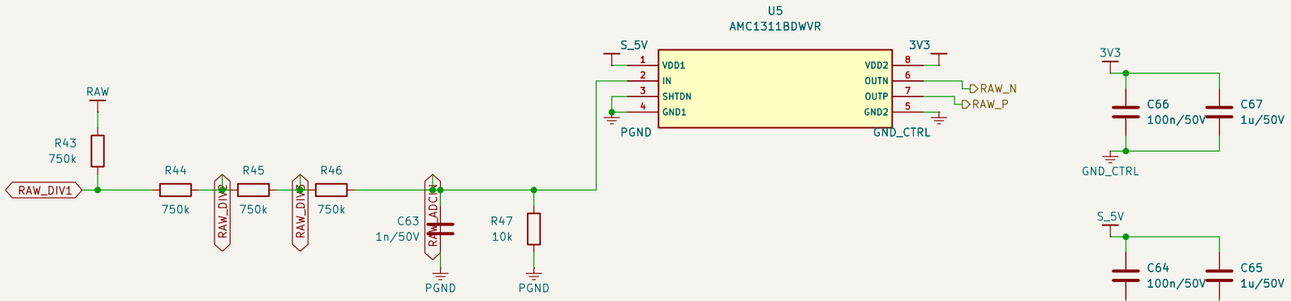
Sheet: /Low-side isolated gate-bias supply/
File: 05_ls_bias.kicad_sch

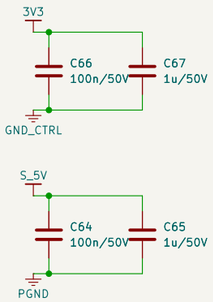
Title: Low-side isolated gate-bias supply

Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 6/9

Raw input, DC–link and output isolated voltage measurement

RAW: independent high–impedance isolated measurement

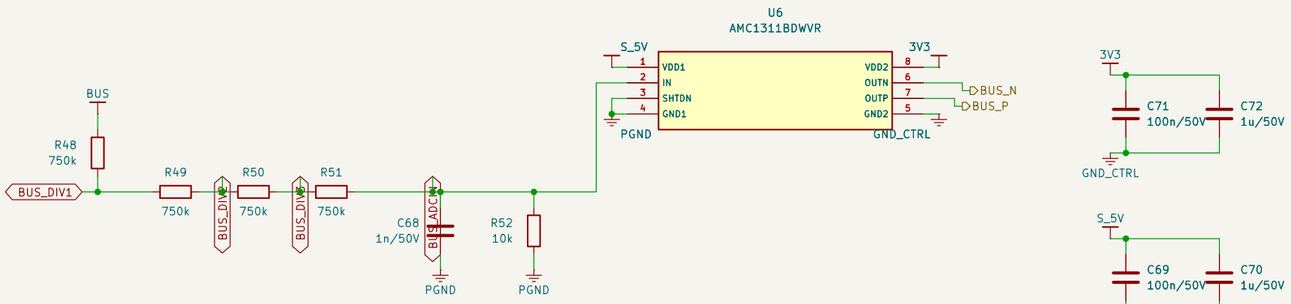


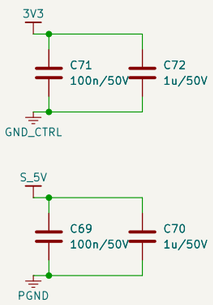


Supply provenance (PWR_FLAG)

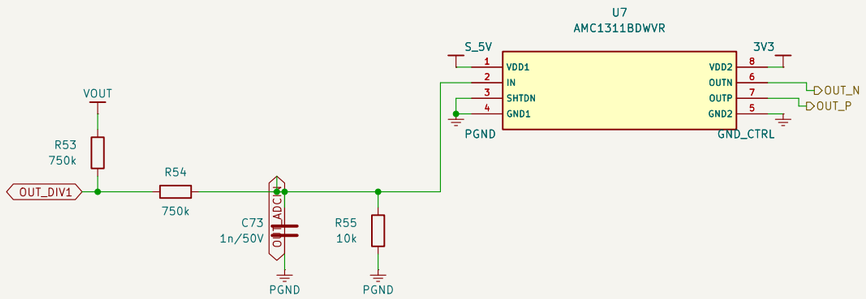


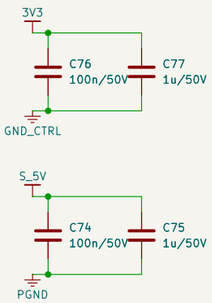
BUS: independent high–impedance isolated measurement





OUT: independent high–impedance isolated measurement





ENGINEERING PROTOTYPE REVIEW – RELEASE BLOCKED – NOT HARDWARE QUALIFIED

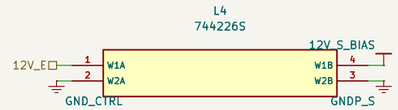
Sheet: /Raw input, DC–link and output isolated voltage measurement/
File: 06_measure.kicad_sch

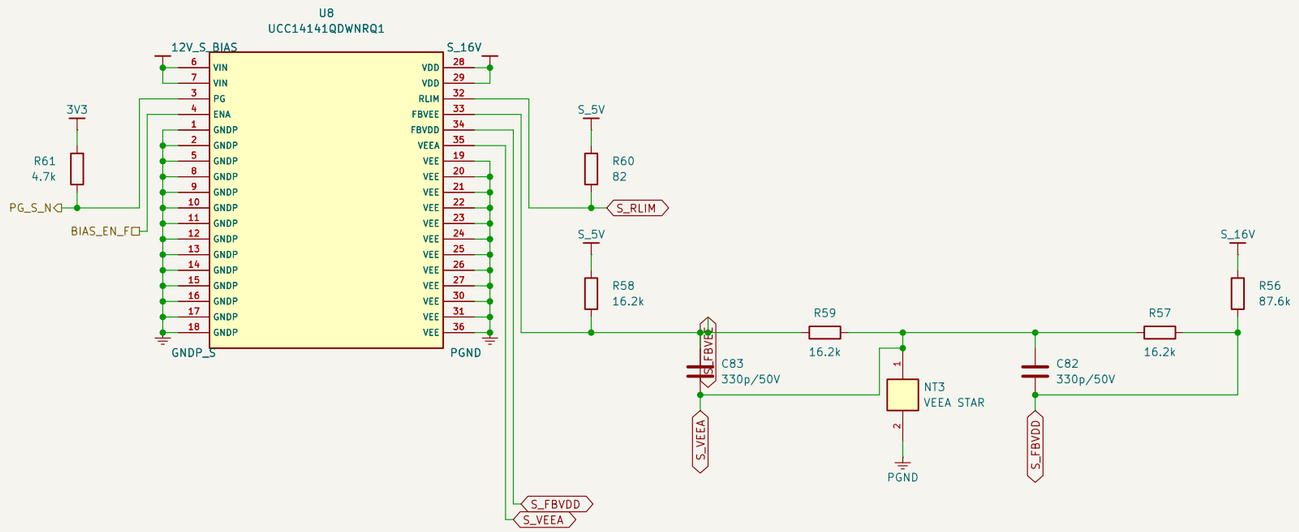
Title: Raw input, DC–link and output isolated voltage measurement

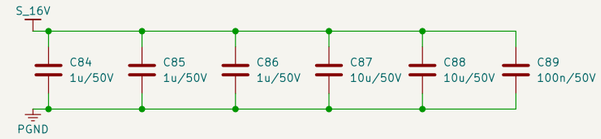
Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 7/9

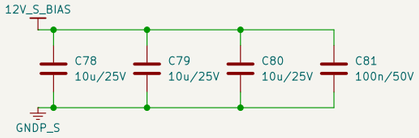
Independent isolated sensing supply

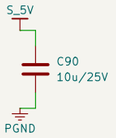
S: isolated DC/DC and physical feedback star



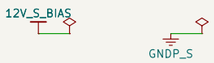








Supply provenance (PWR_FLAG)



ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

Sheet: /Independent isolated sensing supply/
File: 07_sense_bias.kicad_sch

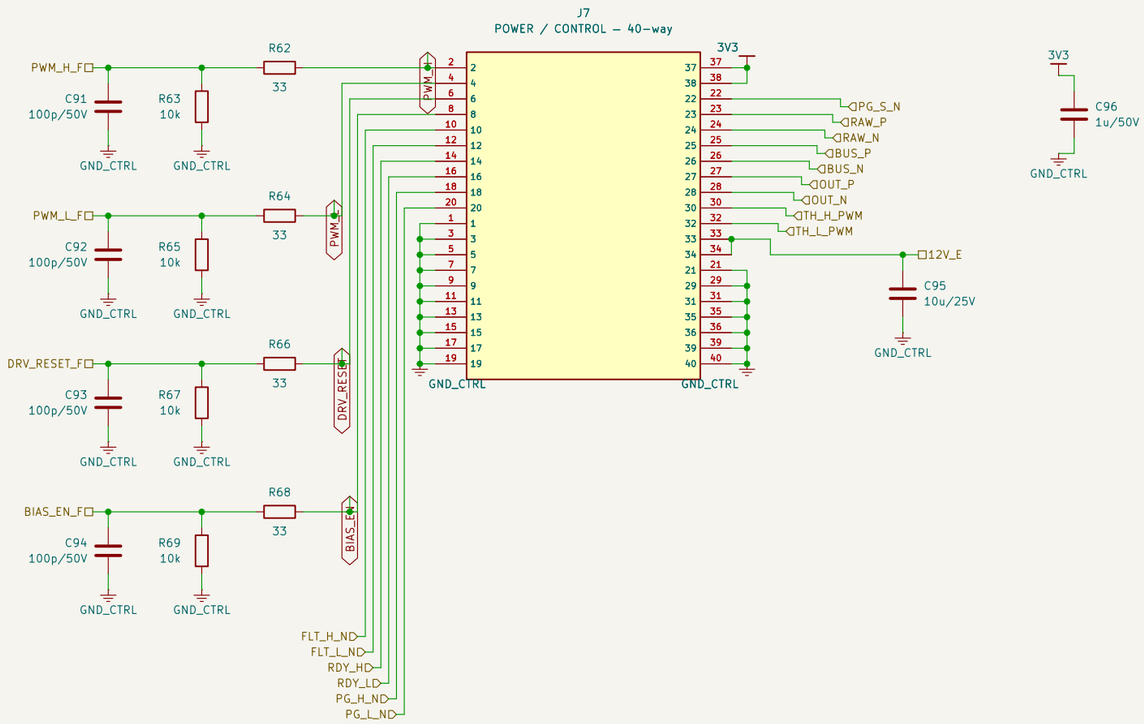
Title: Independent isolated sensing supply

Size: A3	Date:	Rev: V3
KiCad E.D.A. 10.0.7		Id: 8/9

Keyed control harness, fail–safe inputs and mounting

PWR_FLAG symbols state supply provenance for ERC; they do not prove voltage, current capacity or protection.

Keyed SELV cable and fail–safe receive inputs



Mechanical



Supply provenance (PWR_FLAG)



ENGINEERING PROTOTYPE REVIEW – RELEASE BLOCKED – NOT HARDWARE QUALIFIED

Sheet: /Keyed control harness, fail–safe inputs and mounting/
File: 08_interface.kicad_sch

Title: Keyed control harness, fail–safe inputs and mounting

Size: A3	Date:	Rev: V3
KiCad E.D.A. 10.0.7		Id: 9/9